

Economics of *Markets & Decisions*

HSS102: Concepts in
Humanities & Social Sciences

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· Ref: Nicholson & Snyder,
Intermediate Microeconomics (11e)

01 Economic Foundations

What is Economics?

Economics studies how individuals, firms, governments, and other organisations within society make choices, and how those choices determine society's use of its scarce resources. It is fundamentally the science of constrained decision-making.

DEFINITION

Economics is the study of choice under scarcity — how rational agents allocate limited resources among competing uses to maximise welfare.

Three Core Economic Problems

PROBLEM 1

What to produce?

Which goods and services, and in what quantities?

PROBLEM 2

How to produce?

Which inputs, techniques, and technologies to use?

PROBLEM 3

For whom to produce?

Who receives the output? How is it distributed?

Key Economic Concepts

SCARCITY & RESOURCES

Resources (land, labour, capital, entrepreneurship) are finite, but human wants are unlimited. This gap creates scarcity, making choice — and therefore opportunity cost — unavoidable.

ECONOMIC EFFICIENCY

An outcome is economically efficient when resources are allocated such that no reallocation could make anyone better off without making someone else worse off (Pareto efficiency).

Economic Agents

Decisions are made by **economic agents** — entities whose actions affect economic outcomes. There are two main types:

Households

The consuming units in an economy. They own factors of production (labour, capital) and supply them to firms in exchange for income, then use that income to buy goods and services.

Firms

The producing units. They transform inputs (factors of production) into outputs (goods and services) by combining labour, capital, land, and entrepreneurship.

Four Basic Components of an Economy

- 1 Production — Combining inputs to create goods and services
- 2 Consumption — Using goods and services to satisfy wants
- 3 Distribution — Allocating income and output among agents
- 4 Exchange — Trading goods and services through markets

Types of Markets

Markets are classified by the number of sellers, product homogeneity, barriers to entry/exit, and information availability:

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of sellers	Very many	Many	Few	One
Product type	Homogeneous	Differentiated	Homo/Diff	Unique
Entry barriers	None	Low	High	Very high
Price control	None (price taker)	Some	Significant	Full (price maker)
Examples	Agriculture	Restaurants	Telecom, Oil	Railways, Utilities

02 Production Possibilities Curve

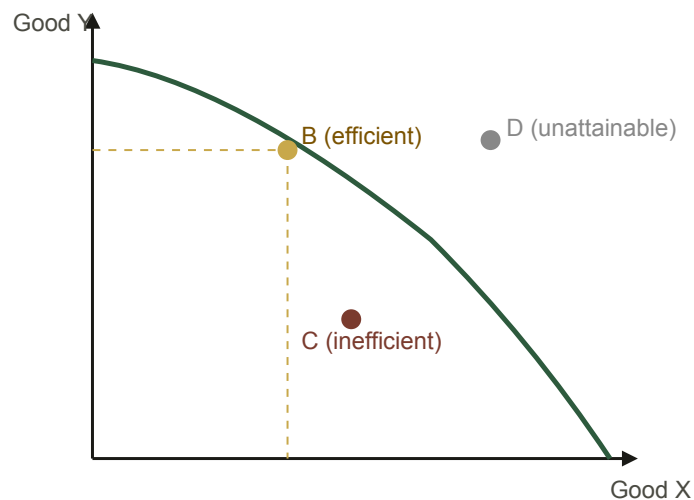
DEFINITION

The Production Possibilities Curve (PPC) shows all possible combinations of two goods an economy can produce with its given resources and technology, operating at full efficiency.

Three Key Assumptions

- 1 The economy operates at full employment and full production capacity (complete efficiency).
- 2 The amount of resources is fixed — no new land, labour, or capital is added.
- 3 The state of technology does not change throughout the period of analysis.

Interpreting the PPC



Points *on* the curve are efficient; *inside* = inefficient; *outside* = unattainable with current resources.

Opportunity Cost on the PPC

Moving along the PPC means trading one good for another. The amount of Good Y sacrificed to gain one more unit of Good X is the opportunity cost. Because of

increasing opportunity costs, the PPC bows outward (is concave to the origin).

Factors that Shift the PPC

OUTWARD SHIFT (GROWTH)

Population ↑

More labour available

OUTWARD SHIFT (GROWTH)

Technology ↑

Same inputs produce more output

OUTWARD SHIFT (GROWTH)

Economic growth

Increased total resources/capital

INWARD SHIFT (CONTRACTION)

Recession / disaster

Resources destroyed or unemployed

03 Theory of Demand

DEFINITION

Demand is the functional relationship between the price of a good and the quantity consumers are willing *and able* to buy in a given period, holding all else constant (*ceteris paribus*). Demand = Willingness to buy + Ability to buy.

The Demand Function

A demand function expresses quantity demanded as a function of price (and other factors). In linear form:

$Q_d = a - bP$ where a = autonomous demand, b = price sensitivity

WORKED EXAMPLE

Given: $Q_d = 6000 - 1000P$. Derive the demand schedule and curve.

PRICE (P)	QUANTITY DEMANDED (QD)
0	6000
1	5000
2	4000
3	3000
4	2000
5	1000
6	0

The inverse demand function (P as a function of Q) is: $P = 6 - Q/1000$. This gives the demand curve a downward slope.

Law of Demand

Consumers demand **more** of a good when its price **falls**, and less when its price rises, holding all other factors constant. This inverse price–quantity relationship holds because of:

Substitution Effect

When a good's price falls, it becomes cheaper relative to substitutes. Consumers switch toward it from other goods.

Income Effect

A price fall increases real purchasing power. Consumers can afford more of all goods, including this one.

ASSUMPTIONS OF THE LAW OF DEMAND

- Tastes and preferences of consumers remain unchanged
- Income of the consumer remains unchanged
- Prices of related goods remain unchanged
- The good should not have prestige/Veblen value

Factors that Influence Demand

A. INTERNAL FACTORS (FIRM-LEVEL)

- Price of the good itself
- Service and payment policies of the seller
- Profit margins and promotional strategies

B.1 ECONOMIC EXTERNAL FACTORS

SUBSTITUTE GOODS

A good that can be used in place of another. If the price of coffee rises, demand for tea increases. The goods are substitutes — they have a *positive* cross-price elasticity.

$$\uparrow P_{\text{coffee}} \rightarrow \uparrow Q_{\text{d tea}}$$

COMPLEMENTARY GOODS

Goods consumed together. If the price of pens rises, demand for ink falls. They have a *negative* cross-price elasticity.

$$\uparrow P_{\text{pen}} \rightarrow \downarrow Q_{\text{d ink}}$$

Normal Goods

Demand *increases* when consumer income rises. Examples: cars, branded clothing, restaurants. Income elasticity is positive.

Inferior Goods

Demand *decreases* when consumer income rises. Examples: public bus transport, low-grade rice, generic brands. Income elasticity is negative.

B.2 NON-ECONOMIC EXTERNAL FACTORS

FACTOR

Tastes & Preferences

Age, gender, culture, education shape demand

FACTOR

Taxation

Higher taxes raise effective price; shift demand

FACTOR

Advertising

Shifts demand curve rightward if effective

FACTOR

Future Price Expectations

Expected price rise → buy now → demand
↑

FACTOR

Festive Seasons

Diwali, holidays boost demand for specific goods

FACTOR

Climate Change

Weather affects demand for ACs, raincoats etc.

Change in Quantity Demanded vs. Change in Demand

Change in Quantity Demanded

Movement *along* the demand curve. Caused only by a change in the price of the good itself. Results in movement from one point to another on the same curve.

Change in Demand

A *shift* of the entire demand curve. Caused by any factor other than the good's own price (income, tastes, related goods' prices, etc.). The whole curve moves left or right.

Exceptional Cases: Upward-Sloping Demand Curves

GIFFEN GOODS (ROBERT GIFFEN)

Essential inferior goods with no close substitutes. When the price rises, the strong negative income effect overwhelms the substitution effect — consumers are now poorer and buy *more* of this staple as they can no longer afford superior alternatives. Example: coarse rice or bread in extreme poverty.

VEBLEN GOODS (THORSTEIN VEBLEN)

Luxury prestige goods where a higher price signals higher status. Higher price increases perceived desirability. The snob effect or Veblen effect makes demand increase with price. Examples: Rolex watches, designer handbags, luxury cars.

Individual vs. Market Demand

Each consumer has their own individual demand function. The market demand is obtained by *horizontal summation* — at every price, add up all individual quantities demanded across all consumers in the market.

HORIZONTAL SUMMATION EXAMPLE

If Consumer A demands 10 units at price ₹5 and Consumer B demands 8 units at ₹5, the market demand at ₹5 = $10 + 8 = 18$ units.

04 Theory of Supply

DEFINITION

Supply is the functional relationship between the price of a good and the quantity producers are willing to sell per period, all else held constant (*ceteris paribus*). A firm is willing to supply more as price rises — so the supply curve slopes upward.

The Supply Function

$Q_s = c + dP$ where c = autonomous supply, d = price responsiveness (positive)

WORKED EXAMPLE

Given: $Q_s = 1000P$. Derive the supply schedule.

PRICE (P)	QUANTITY SUPPLIED (Q _S)
1	1000
2	2000
3	3000
4	4000

Factors that Shift the Supply Curve

STATE OF TECHNOLOGY

Improved technology reduces production costs, allowing more output at every price. Supply shifts rightward. E.g., introduction of automated machinery.

INPUT PRICES

Higher wages, raw material costs, or capital costs raise production costs. Supply shifts leftward. Lower input costs shift it rightward.

PRICES OF RELATED GOODS IN PRODUCTION

If wheat and cotton use the same farmland, a rise in wheat prices may divert supply away from cotton (substitutes in production). Supply of cotton shifts left.

FUTURE EXPECTATIONS

If producers expect prices to rise next month, they may withhold current supply to sell more later. Current supply decreases.

FACTOR

Number of Producers

FACTOR

Government Regulations

More firms → market supply increases

Taxes reduce supply; subsidies increase it

FACTOR

Externalities

Pollution costs increase effective cost of supply

Change in Quantity Supplied vs. Change in Supply

Change in Quantity Supplied

Movement *along* the supply curve due to a change in the own price of the good.

No shift occurs.

Change in Supply

A *shift* of the entire supply curve due to changes in technology, input prices, number of firms, etc. — i.e., any determinant other than own price.

05 Market Equilibrium

EQUILIBRIUM PRICE

The price at which $Q_d = Q_s$ — the market clears, with no surplus or shortage. This is the price the market gravitates toward and rests at.

Finding Equilibrium Algebraically

Set $Q_d = Q_s$ and solve for the equilibrium price (P^*), then substitute back to find equilibrium quantity (Q^*).

WORKED EXAMPLE 1

$$Q_d = 500 - 10P \quad \text{and} \quad Q_s = 50 + 5P$$

$$500 - 10P = 50 + 5P \quad 450 = 15P \quad P^* = 30 \quad Q^* = 500 - 10(30) = 200$$

Equilibrium: $P^* = ₹30$, $Q^* = 200$ units

WORKED EXAMPLE 2

$$Q_d = 160 - 40P \quad \text{and} \quad Q_s = 50 + 15P$$

$$160 - 40P = 50 + 15P \quad 110 = 55P \quad P^* = 2 \quad Q^* = 160 - 40(2) = 80$$

Equilibrium: $P^* = ₹2$, $Q^* = 80$ units

Shocks to Equilibrium

DEMAND INCREASES (RIGHTWARD SHIFT)

$$P^* \uparrow \quad Q^* \uparrow$$

E.g., income rises, population grows

DEMAND DECREASES (LEFTWARD SHIFT)

$$P^* \downarrow \quad Q^* \downarrow$$

E.g., substitute becomes cheaper

SUPPLY INCREASES (RIGHTWARD SHIFT)

$$P^* \downarrow \quad Q^* \uparrow$$

E.g., technology improves, input costs fall

SUPPLY DECREASES (LEFTWARD SHIFT)

$$P^* \uparrow \quad Q^* \downarrow$$

E.g., input prices rise, natural disaster

Utility Concepts

TOTAL UTILITY (TU)

Total satisfaction

Aggregate utility from all units consumed

MARGINAL UTILITY (MU)

$$\Delta TU / \Delta Q$$

Extra utility from consuming one more unit

LAW OF DIMINISHING MU

MU falls as Q rises

Each additional unit adds less satisfaction than the last

Indifference Curves (IC)

DEFINITION

An Indifference Curve shows all combinations of two goods that give a consumer the *same level of utility*. The consumer is indifferent between any point on the same curve.

PROPERTIES OF INDIFFERENCE CURVES (FROM NICHOLSON & SNYDER)

- 1 Downward sloping — to maintain constant utility, if you consume more of X, you must consume less of Y.
- 2 Cannot intersect — if two ICs crossed, the intersection point would give two different utility levels simultaneously, which is a contradiction.
- 3 Convex to origin — reflects the diminishing marginal rate of substitution: as you gain more X, you are willing to give up less and less Y to get still more X.
- 4 Higher IC = higher utility — a curve farther from the origin represents more of both goods and thus greater satisfaction.

Marginal Rate of Substitution (MRS)

The MRS is the rate at which a consumer is willing to exchange one good for another while remaining at the same utility level. It equals the (negative) slope of the indifference curve.

$$MRS_{XY} = -\Delta Y / \Delta X = MU_X / MU_Y$$

As you move *down* the indifference curve (gaining X, sacrificing Y), the MRS diminishes — you become less and less willing to give up Y for more X. This is the principle of Diminishing MRS, which explains why ICs are convex.

Special Cases of Indifference Curves

Perfect Substitutes

ICs are straight lines with constant slope. The consumer is always willing to trade one good for the other at a fixed rate (e.g., Exxon vs. Chevron petrol). MRS is constant.

Perfect Complements

ICs are L-shaped. Goods must be consumed in fixed proportions (e.g., left shoe and right shoe). No utility is gained from having more of just one.

07 Budget Constraint & Optimal Choice

The Budget Constraint

DEFINITION

The budget constraint shows all combinations of two goods a consumer can afford given their income (I) and the prices of the goods (P_x, P_y).

$P_x \cdot X + P_y \cdot Y = I$ (all income is spent) Solving for Y : $Y = I/P_y - (P_x/P_y) \cdot X$ Slope = $-P_x/P_y$ (opportunity cost of X in terms of Y)

SHIFTS IN THE BUDGET LINE

Income Increases

Budget line shifts outward parallel — the consumer can now afford more of both goods. The slope (price ratio) does not change.

Price of X Rises

Budget line rotates inward on the X -axis — the X -intercept (I/P_x) falls. The Y -intercept stays the same. The slope becomes steeper.

Utility Maximisation — The Optimal Choice

The consumer maximises utility by choosing the bundle where the highest attainable indifference curve is tangent to the budget line. At this point:

$MRS = P_x / P_y$ The subjective trade-off rate = the market trade-off rate Equivalently: $MU_x / P_x = MU_y / P_y$ Marginal utility per rupee is equal across all goods

Intuition (Nicholson & Snyder)

If $MRS > P_x/P_y$, the consumer values X more than the market does — they should buy more X (and less Y) until MRS falls to the price ratio. If $MRS < P_x/P_y$,

the reverse applies. Only at $MRS = P_x/P_y$ is there no further utility-improving reallocation possible.

NUMERICAL EXAMPLE

A student has ₹30. Hamburgers cost ₹3 (X); soft drinks cost ₹1.50 (Y).

Budget: $3X + 1.5Y = 30 \rightarrow Y = 20 - 2X$ Slope = $-P_x/P_y = -3/1.5 = -2$ One more hamburger costs 2 soft drinks (opportunity cost)

Optimal point: choose X^*, Y^* where $MRS = 2$ (the price ratio). Beyond this, no reallocation improves utility.

08 Theory of Production

PRODUCTION FUNCTION

$Q = f(L, K)$ — shows the maximum output (Q) obtainable from combinations of labour (L) and capital (K), given the state of technology.

Short Run vs. Long Run

Short Run

At least one input is fixed (typically capital). Only variable inputs can be changed. Firms face the Law of Variable Proportions.

Long Run

All inputs are variable. Firms can adjust capital as well as labour. Returns to scale apply.

Law of Variable Proportions (One Variable — L or K)

As variable input (labour) is added to a fixed input (capital), output first rises at an increasing rate, then at a decreasing rate (diminishing returns), and may eventually fall.

TOTAL PRODUCT (TP)

Total output from all units of L

AVERAGE PRODUCT (AP)

TP / L

Output per worker

MARGINAL PRODUCT (MP)

$\Delta TP / \Delta L$

Extra output from one more worker

Law of Diminishing Marginal Returns

Holding capital fixed, as more workers are added, the marginal product of labour eventually falls. This is the short-run constraint on production. Mathematically, MP reaches its maximum before AP does; and $AP = MP$ at the peak of AP.

Two-Variable Production — Isoquants

ISOQUANT

A curve showing all (L, K) combinations that produce the *same quantity of output*. Analogous to an indifference curve but for the firm's production choices.

MARGINAL RATE OF TECHNICAL SUBSTITUTION (MRTS)

$MRTS_{LK} = -\Delta K / \Delta L = MP_L / MP_K$ Rate at which capital can be replaced by labour holding output constant

MRTS diminishes as you move down the isoquant — as labour replaces capital, each additional unit of labour becomes less productive (and vice versa), so more labour is needed to replace each unit of capital.

Isocost Line & Firm Equilibrium

The isocost line shows all (L, K) combinations that cost the same total amount. Analogous to the consumer's budget line.

$wL + rK = C$ where w = wage rate, r = rental cost of capital, C = total cost Slope = $-w/r$

The firm minimises cost (or maximises output) at the tangency point between the isocost line and the isoquant: $MRTS = w/r$.

Returns to Scale

In the long run, when *all* inputs are scaled by factor λ :

INCREASING RETURNS TO SCALE

$$\text{Output} > \lambda \times Q$$

Doubling inputs more than doubles output.
Economies of scale.

CONSTANT RETURNS TO SCALE

$$\text{Output} = \lambda \times Q$$

Doubling inputs exactly doubles output.

DECREASING RETURNS TO SCALE

$$\text{Output} < \lambda \times Q$$

Doubling inputs less than doubles output.
Diseconomies of scale.

Types of Cost

ACCOUNTING COST

Explicit, out-of-pocket payments made for inputs: wages, rent, raw materials, utilities. Recorded in financial statements.

ECONOMIC COST

Accounting cost plus implicit/opportunity costs. A firm using its own building forgoes rental income — this is an economic cost even if not an accounting one.

OPPORTUNITY COST

The value of the *next best forgone alternative*. The true cost of any choice in economics. Even "free" resources have opportunity cost.

SUNK COST

Already-incurred costs that *cannot be recovered*. Rational economic decisions should ignore sunk costs and focus only on future costs and benefits.

Cost Functions

$$TC = TFC + TVC \quad ATC = TC / Q = AFC + AVC \quad MC = \Delta TC / \Delta Q \quad AFC =$$

TFC / Q (always falls as Q rises)

COST TABLE EXAMPLE (TFC = ₹50 FIXED)

OUTPUT (Q)	TFC	TVC	TC	AFC	AVC	ATC	MC
0	50	0	50	—	—	—	—
1	50	50	100	50.00	50.00	100.00	50
2	50	78	128	25.00	39.00	64.00	28
3	50	98	148	16.67	32.67	49.33	20
4	50	112	162	12.50	28.00	40.50	14
5	50	130	180	10.00	26.00	36.00	18
6	50	150	200	8.33	25.00	33.33	20
7	50	175	225	7.14	25.00	32.14	25
8	50	204	254	6.25	25.50	31.75	29
9	50	242	292	5.56	26.89	32.44	38
10	50	300	350	5.00	30.00	35.00	58
11	50	385	435	4.55	35.00	39.55	85

Key Relationships

MC and the U-Shaped ATC

AFC falls continuously. AVC is U-shaped (first falls due to specialisation, then rises due to diminishing returns). $ATC = AFC + AVC$ is also U-shaped. The MC

curve passes through the minimum points of both AVC and ATC. When $MC < ATC$, ATC is falling; when $MC > ATC$, ATC is rising.

10 Elasticity

Price Elasticity of Demand ($|e_d|$)

DEFINITION

Price elasticity of demand measures the percentage change in quantity demanded in response to a 1% change in price. It tells us how sensitive consumers are to price changes.

$$|e_d| = \left| \% \Delta Q_d / \% \Delta P \right| = \left| (\Delta Q / \Delta P) \times (P / Q) \right|$$

DEGREES OF PRICE ELASTICITY

$$|e_d| = 0$$

Perfectly Inelastic

Vertical demand curve. Q unresponsive to P. E.g., life-saving medicine.

$$|e_d| < 1$$

Inelastic

Q changes less than P. E.g., petrol, cigarettes, salt.

$$|e_d| = 1$$

Unitary Elastic

% change in Q = % change in P. Total expenditure unchanged.

$$|e_d| > 1$$

Elastic

Q changes more than P. E.g., luxury goods, airline tickets.

$$|e_d| = \infty$$

Perfectly Elastic

Horizontal demand curve. Any price rise $\rightarrow$ zero demand.

Arc Price Elasticity

Measures elasticity over a range of prices (between two points) using midpoint averages to avoid direction-of-change bias:

$$\text{Arc } |e_d| = [(Q_2 - Q_1) / ((Q_1 + Q_2) / 2)] \div [(P_2 - P_1) / ((P_1 + P_2) / 2)]$$

Income Elasticity of Demand (e_Y)

$$e_Y = \% \Delta Q_d / \% \Delta \text{Income}$$

$$EY > 1$$

Luxury Good

Demand rises proportionally more than income

$$0 < EY < 1$$

Normal Good (necessity)

Demand rises but less than proportionally

$$EY < 0$$

Inferior Good

Demand falls as income rises

Cross-Price Elasticity of Demand (e_{XY})

$$e_{XY} = \% \Delta Q_{dX} / \% \Delta P_Y$$

$$e_{XY} > 0 \rightarrow \text{Substitutes}$$

Price of Y rises, demand for X rises. Tea and coffee; Pepsi and Coke.

$$e_{XY} < 0 \rightarrow \text{Complements}$$

Price of Y rises, demand for X falls. Car and petrol; pen and ink.

Consumer & Producer Surplus

RESERVATION PRICE

The maximum price a consumer is willing to pay for a unit. When the market price is below this, the consumer gains a surplus.

CONSUMER SURPLUS

The difference between what consumers are *willing to pay* (reservation price) and what they *actually pay* (market price). Graphically, it is the area below the demand curve and above the market price.

$$CS = \text{Willingness to Pay} - \text{Price Paid}$$

PRODUCER SURPLUS

The difference between the *market price* received and the *minimum price* the producer needed to supply each unit. Graphically, it is the area above the supply curve and below the market price.

$$PS = \text{Price Received} - \text{Minimum Acceptable Price}$$

Total Surplus & Economic Efficiency

Total surplus = CS + PS. In a competitive market, total surplus is maximised at the equilibrium price — this is why perfectly competitive markets are considered economically efficient. Any price control (price floor or ceiling) away from equilibrium reduces total surplus, creating deadweight loss.

12 Perfect Competition

Assumptions

- 1 Many buyers and sellers — no individual can influence price
- 2 Homogeneous products — goods are perfectly substitutable across firms
- 3 Free entry and exit — no barriers prevent firms entering or leaving
- 4 Price taking — each firm takes market price as given; $P = MR = AR$
- 5 Profit maximisation — firms aim to maximise π
- 6 Perfect information — all agents know all prices and technologies
- 7 Perfect factor mobility — inputs can freely move to highest-return use

Revenue Under Perfect Competition

Since the firm is a price taker, it faces a perfectly elastic (horizontal) demand curve at the market price P .

$P = MR = AR$ Price = Marginal Revenue = Average Revenue (all equal market price)
 $TR = P \times Q$ $AR = TR/Q = P$ $MR = \Delta TR / \Delta Q = P$

Short-Run Profit Maximisation

A firm maximises profit by producing where $MR = MC$, i.e., $P = MC$. The profit or loss depends on the relationship between P and ATC :

$$P > ATC$$

Supernormal profit

Firm earns economic profit

$$P = ATC$$

Break-even (normal profit)

$TR = TC$; zero economic profit

$$AVC \leq P < ATC$$

Loss, but continue

Covers variable costs; loss < TFC

$$P < AVC$$

Shut down

Cannot cover variable costs; exit in short run

Long-Run Equilibrium

In the long run, supernormal profits attract entry, driving down price until $P = ATC$ (zero economic profit). Losses cause exit, raising price back to $P = ATC$. The long-run equilibrium condition is:

$$P = MR = MC = ATC \text{ (at the minimum of ATC)}$$

This means resources are used most efficiently — productive efficiency (min ATC) and allocative efficiency ($P = MC$) are both achieved.

13 Monopoly

Assumptions & Definition

MONOPOLY

A market structure with a single seller who produces a good with no close substitutes, protected by barriers to entry. The monopolist is a price maker — it faces the entire downward-sloping market demand curve.

Causes of Monopoly

- 1 Ownership of strategic raw materials or exclusive production knowledge
- 2 Patent rights for a product or production process

- 3 Government licensing or trade barriers excluding competitors
- 4 Market size too small to support more than one optimally-sized plant
- 5 Natural monopoly — substantial economies of scale requiring only one plant
- 6 Government-owned utilities to prevent consumer exploitation

Revenue Under Monopoly

Because the monopolist faces a downward-sloping demand curve, it must lower price to sell more. This means $MR < P$ at every output level.

Since $P = a - bQ$ (demand): $TR = P \times Q = (a - bQ) \times Q = aQ - bQ^2$
 $MR = \Delta TR / \Delta Q = a - 2bQ$
MR has the same intercept as demand but twice the slope
Always: $MR < P$ (MR lies below the demand curve)

Profit Maximisation Under Monopoly

The monopolist still maximises profit where $MR = MC$, then sets price from the *demand curve* at that output level. Since $P > MR = MC$, monopoly price exceeds marginal cost — causing allocative inefficiency.

Deadweight Loss of Monopoly

The monopolist restricts output below the socially optimal level (where $P = MC$ in perfect competition). This creates a deadweight loss — a permanent welfare loss to society. The monopolist's gain comes partly at consumers' expense (transfer) and partly from this destroyed value.

Measuring Monopoly Power — Lerner Index (L)

$$L = (P - MC) / P = -1 / e_d$$

L ranges from 0 (perfect competition; $P = MC$) to approaching 1 (strong monopoly). A higher L indicates greater market power. The index shows that a firm can charge a higher markup when demand is *more inelastic* (smaller $|ed|$).

Rule of Thumb for Monopoly Pricing

$$P = MC / (1 + 1/e_d) \text{ Lower } |ed| \rightarrow \text{higher price relative to MC}$$

SHORT-RUN MONOPOLY PROFIT EXAMPLE

If a monopolist faces Demand: $P = 100 - Q$ and $MC = 20$, find profit-maximising Q and P.

$$\begin{aligned} MR &= 100 - 2Q \text{ Set } MR = MC: 100 - 2Q = 20 \rightarrow Q^* = 40 \\ P^* &= 100 - 40 = ₹60 \text{ Profit per unit} = P - MC = 60 - 20 = ₹40 \\ &\text{(if } MC = ATC \text{ here) Total Profit} = 40 \times 40 = ₹1,600 \end{aligned}$$

14 Imperfect Competition

Imperfect competition exists whenever individual sellers can affect the price of their output. It includes monopoly, oligopoly, and monopolistic competition.

Summary Comparison

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Sellers	Very many	Many	Few (2–10)	One

Product	Identical	Differentiated	Similar or diff.	Unique, no substitutes
Entry barriers	None	Low	Significant	Very high
Price control	None	Limited	Moderate–high	Full (price maker)
MR vs P	$MR = P$	$MR < P$	$MR < P$	$MR < P$
Long-run profit	Zero (normal)	Zero (entry)	Positive possible	Positive (barriers)
Efficiency	Allocatively efficient	Allocatively inefficient	Varies	Allocatively inefficient

Monopolistic Competition

A large number of sellers offer **differentiated** but similar products (restaurants, clothing brands, shampoos). Each has a small degree of price-setting power due to product differentiation, but free entry drives economic profit to zero in the long run. The long-run outcome:

$P = ATC$ but $P > MC$ (unlike perfect competition)

Oligopoly

A few dominant firms that are highly interdependent. Each firm's decisions on price and output significantly affect rivals, leading to **strategic behaviour** (studied with Game Theory). Key features:

- Price rigidity (kinked demand curve model)
- Possibility of collusion (cartels) or price wars
- Non-price competition: advertising, product differentiation
- Barriers to entry protect existing firms' profits

Monopsony

MONOPSONY

A market with many sellers but only one buyer. The single buyer has market power on the demand side, analogous to monopoly on the supply side. Example: a company as the sole employer in a small town. The monopsonist pays wages below the competitive level.

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HSS102 Economics Notes · Based on: Nicholson & Snyder, *Intermediate Microeconomics and Its Application*, 11th Edition (2010) · Also references: Mankiw, *Principles of Microeconomics*; Pindyck & Rubinfeld, *Microeconomics* · March–April 2026